ECE 340: Semiconductor Devices

Lecture 18: Diffusion and Recombination & Fabrication fundamentals

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Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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Diffusion Length: Steady State

Steady-State Diffusion Equations

$$\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}$$

In steady state, $\frac{\partial \delta p}{\partial t} = 0$

$$\therefore\ D_p \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{\tau_p} \text{ or } \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{D_p \tau_p} \equiv \frac{\delta p}{L_p^2}$$

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}$$

In steady state, $\frac{\partial \delta n}{\partial t} = 0$

$$\therefore\ D_n \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{\tau_n} \text{ or } \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{D_n \tau_n} \equiv \frac{\delta n}{L_n^2}$$

Hole Injection at x=0



(The injection mechanism is not specified.)

L_p is the average distance a hole travels before recombining.
L_n is the average distance an electron travels before recombining.

$$L_n \equiv \sqrt{D_n \tau_n} \text{ and } L_p \equiv \sqrt{D_p \tau_p}$$

Excess Hole Distribution and Current

Solution to Hole Diffusion Equation

$$\frac{d^2\delta p}{dx^2} = \frac{\delta p}{L_p^2} \qquad \frac{\partial \delta p}{\partial t} = 0$$

Solution:

$$\delta p(x) = C_1 e^{x/L_p} + C_2 e^{-x/L_p}$$

Boundary Condition:

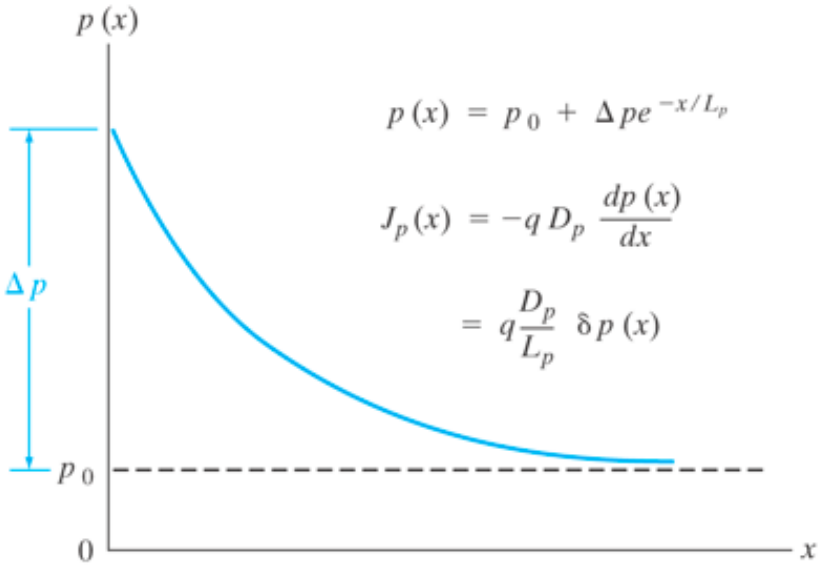
$$\delta p(x=0) = \Delta p = C_1 + C_2$$

$$\delta p(x=\infty) = 0 = C_1 e^{x/L_p}$$

$$C_1 = 0, \quad C_2 = \Delta p$$

$$\delta p(x) = \Delta p \, e^{-x/L_p}$$

Hole Injection at x=0



$$J_p(x) = -qD_p \frac{dp}{dx} = -qD_p \frac{d(p_o + \delta p)}{dx} = -qD_p \frac{d\delta p}{dx}$$

$$= q\frac{D_p}{L_p}\Delta p e^{-x/L_p} = q\frac{D_p}{L_p}\delta p(x)$$

Diffusion Length: Statistical Interpretation

Probability A Hole Injected at x = 0 Survives to "x":

$$\frac{\delta p(x)}{\Delta p} = \frac{\text{Number of Excess Holes at "x"}}{\text{Number of Excess Holes at x=0}} = \frac{\Delta p e^{-x/L_p}}{\Delta p} = e^{-x/L_p}$$

Probability A Hole Will Recombine in the Interval dx :

$$\frac{\text{Holes Lost in Interval dx}}{\text{Number of Excess Holes at x}} = \frac{\delta p(x) - \delta p(x+dx)}{\delta p(x)} = \frac{-\left(d\delta p(x)/dx\right)dx}{\delta p(x)} = \frac{-\left(-\frac{1}{L_p}\right)\Delta p e^{-x/L_p}dx}{\Delta p e^{-x/L_p}} = \frac{1}{L_p}dx$$

The probability a hole will recombine at "x" equals the probablility it survives to "x" times the probability it recombines at "x":

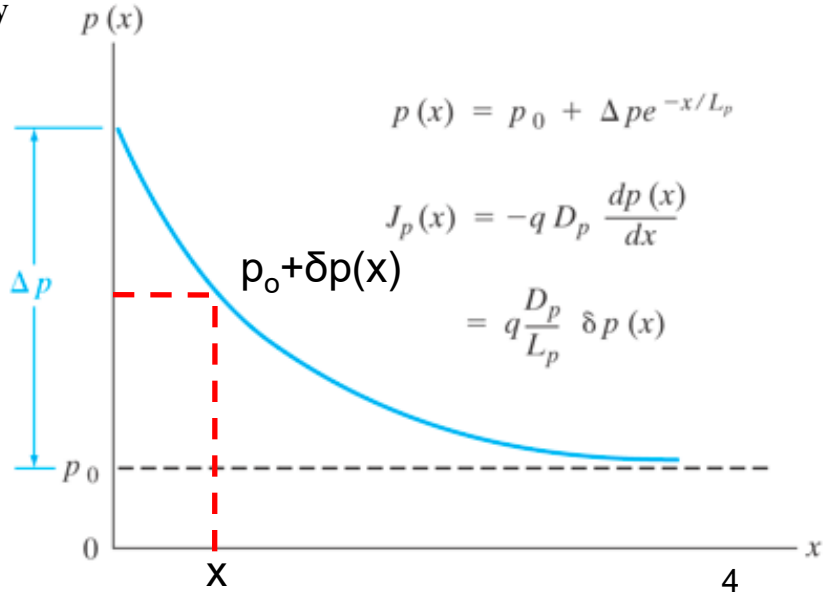
$$\text{P(hole recombination at "x")} = \left(e^{-x/L_p}\right)\left(\frac{1}{L_p}dx\right) = \frac{1}{L_p}e^{-x/L_p}dx$$

But recall the definition of the expectation value of a function:

$$\langle f(x)\rangle = \int_{-\infty}^{\infty} f(x)P(x)dx$$

so for the function $f(x) = x$:

$$\langle x\rangle = \int_0^{\infty} x\frac{1}{L_p}e^{-x/L_p}dx = L_p$$



Current Flow and Charge Loss

$$\delta p(x) = \Delta p \bullet e^{-x/L_p}$$

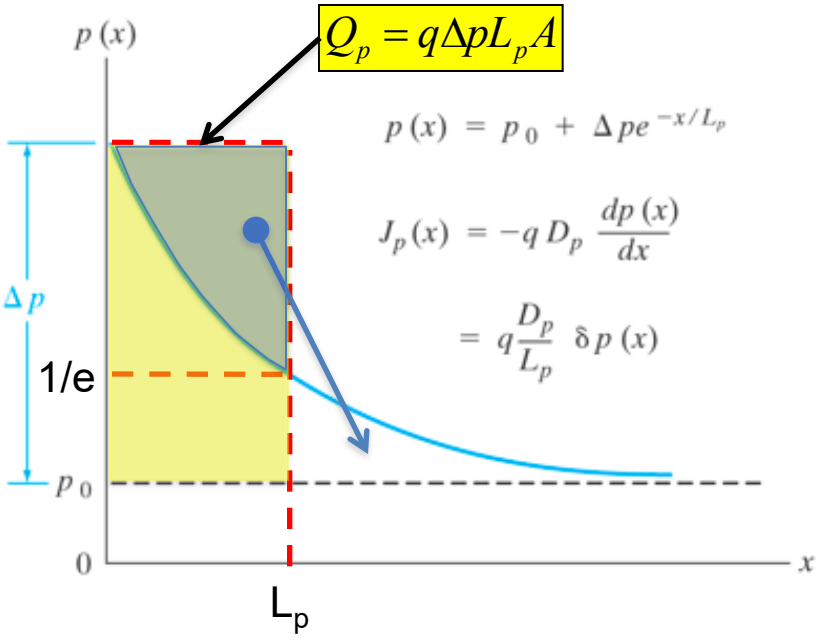
Note, the total number of excess carriers is:

$$A \times \int_0^\infty \Delta p e^{-x/L_p}\, dx = -A\Delta p L_p e^{-x/L_p}\Big|_0^\infty = -A\Delta p L_p \left(e^{-\infty/L_p} - e^0\right)$$

$$= -A\Delta p L_p \left(e^{-\infty/L_p} - e^0\right)$$

$$= A\Delta p L_p \quad \leftarrow \text{Total excess carrier}$$

$$\boxed{Q_p = q\Delta p L_p A} \quad \text{Total stored charge}$$



Hole current J_p is:

$$J_p(x) = -qD_p \frac{dp(x)}{dx} = -qD_p \frac{d\delta p(x)}{dx} = q\frac{D_p}{L_p}\delta p(x)$$

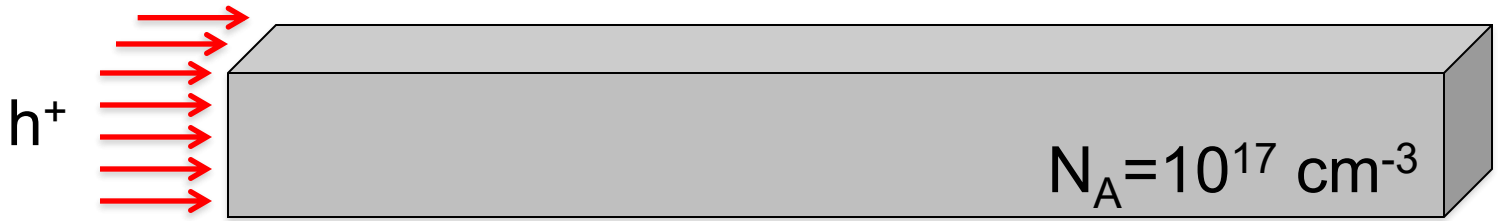
$$J_p(0) = q\frac{D_p}{L_p}\Delta p = q\left(\frac{L_p^2}{\tau_p}\right)\frac{1}{L_p}\Delta p = \frac{qL_p\Delta pA}{\tau_p A} = \frac{Q_p}{\tau_p A}$$

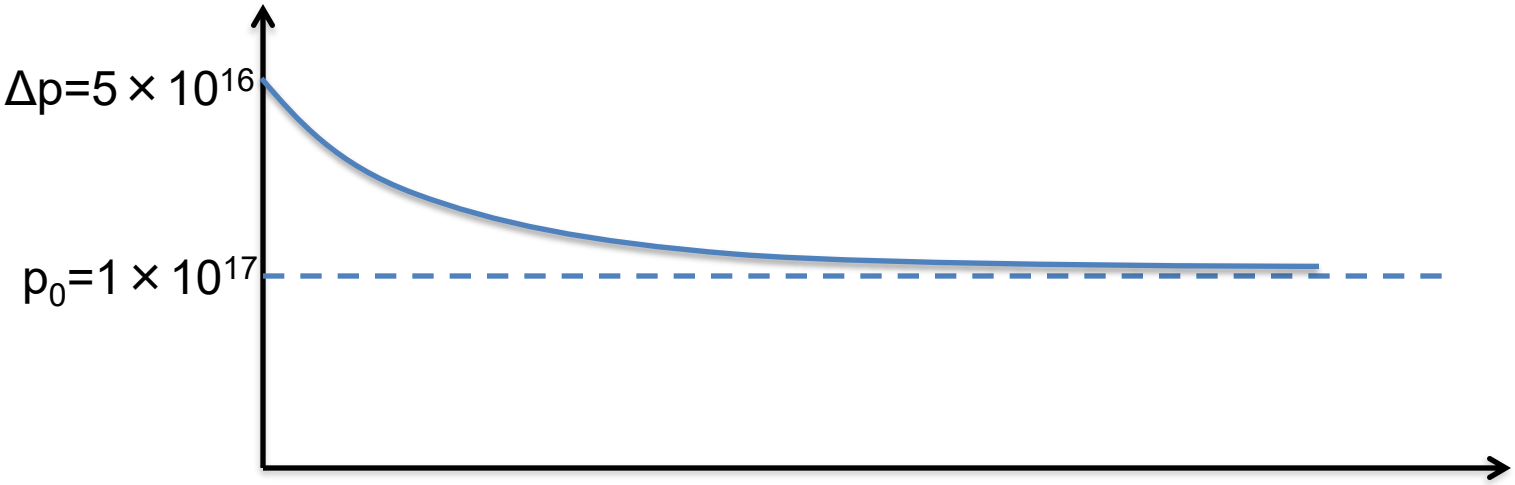
$$J_p(0) = \frac{Q_p}{\tau_p A} \quad , \quad I_p(0) = \frac{Q_p}{\tau_p}$$

- The current flow into the device at x=0 is equal to the charge loss due to recombination

Example

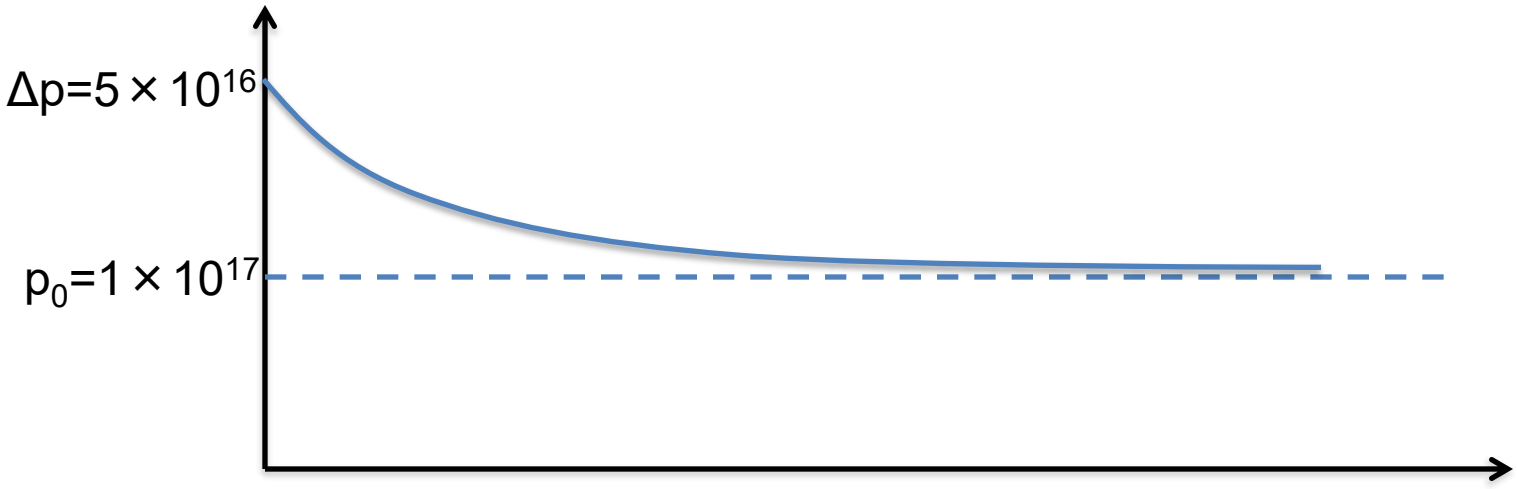
- In a long p-type Si bar with a cross sectional area of 0.5 cm^2 and $N_a=10^{17}$ cm^{-3}, we inject holes such that the steady state excess hole concentration is $5x10^{16}$ cm^{-3} at x=0.





Example

- In a long p-type Si bar with a cross sectional area of 0.5 cm^2 and $N_a=10^{17}$ cm^{-3}, we inject holes such that the steady state excess hole concentration is $5x10^{16}$ cm^{-3} at x=0. (1) What is the hole concentration at x=1000Å? (2) What is the hole current there? (3) How much is the excess stored hole charge? Assume that $\mu_p=500$ $cm^2/V\text{-}s$ and $\tau_p=10^{-10}s$, T=300 K.



Example (2)

- What is the hole concentration at x=1000Å? Assume that μ_p=500 cm^2/V-s and τ_p =10^{-10}s, T=300 K.

$$D_p = \frac{kT}{q}\mu_p = 0.0259 \times 500 = 12.95 \text{ cm}^2/\text{s}$$ ← $\frac{D}{\mu} = \frac{kT}{q}$

$$L_p = \sqrt{D_p \tau_p} = \sqrt{12.95 \times 10^{-10}} = 3.6 \times 10^{-5} \text{ cm}$$

$p(x) = p_0 + \delta p(x)$
$= p_0 + \Delta p e^{-x/L_p}$

$$p(x) = p_o + \Delta p e^{-x/L_p} = 10^{17} + 5 \times 10^{16} e^{-\frac{10^{-5}}{3.6 \times 10^{-5}}}$$
$$= 1.379 \times 10^{17}$$

Example (3)

- What is the hole current there?
- How much is the excess stored hole charge in the bar?

$$J_p(x) = -qD_p \frac{dp(x)}{dx} \rightarrow I_p = -qAD_p \frac{dp}{dx} = qA\frac{D_p}{L_p}(\Delta p)e^{-x/L_p}$$

$$= 1.6\times10^{-19} \times 0.5 \times \frac{12.95}{3.6\times10^{-5}} \times 5\times10^{16} e^{-\frac{10^{-5}}{3.6\times10^{-5}}}$$

$$= 1.09\times10^{3} \text{ A}$$

$$\mathrm{Q}_p = qA(\Delta p)L_p$$

$$= 1.6\times10^{-19}(0.5)\left(5\times10^{16}\right)\left(3.6\times10^{-5}\right)$$

$$= 1.44\times10^{-7} \text{ C}$$

Instructional Objectives (1)

By the time of exam No. 1 (after 17 lectures), the students should be able to do the following:

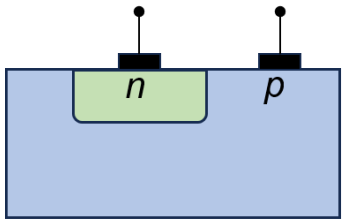
1. Outline the classification of solids as metals, semiconductors, and insulators and distinguish direct and indirect semiconductors.
2. Determine relative magnitudes of the effective mass of electrons and holes from an E(k) diagram.
3. Calculate the carrier concentration in intrinsic semiconductors.
4. Apply the Fermi-Dirac distribution function to determine the occupation of electron and hole states in a semiconductor.
5. Calculate the electron and hole concentrations if the Fermi level is given; determine the Fermi level in a semiconductor if the carrier concentration is given.
6. Determine the variation of electron and hole mobility in a semiconductor with temperature, impurity concentration, and electrical field.
7. Apply the concept of compensation and space charge neutrality to calculate the electron and hole concentrations in compensated semiconductor samples.
8. Determine the current density and resistivity from given carrier densities and mobilities.
9. Calculate the recombination characteristics and excess carrier concentrations as a function of time for both low level and high level injection conditions in a semiconductor.
10. Use quasi-Fermi levels to calculate the non-equilibrium concentrations of electrons and holes in a semiconductor under uniform photoexcitation.
11. Calculate the drift and diffusion components of electron and hole currents.
12. Calculate the diffusion coefficients from given values of carrier mobility through the Einstein's relationship and determine the built-in field in a non-uniformly doped sample.
13. Calculated diffusion current for steady-state carrier injection

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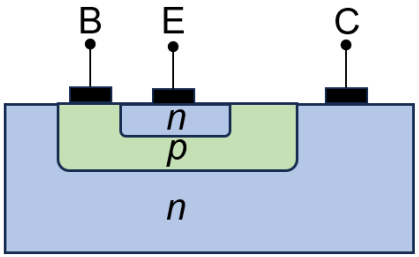
Fabrication Fundamentals

Fundamental devices

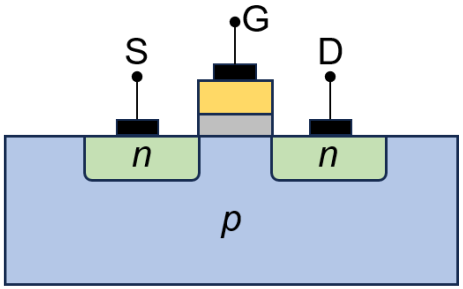
P-N diode



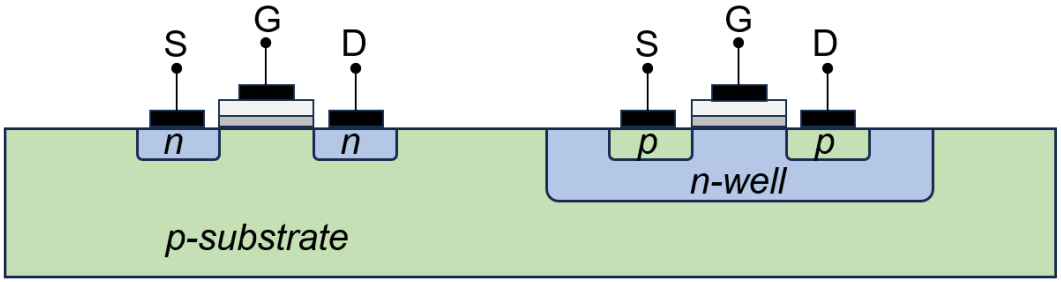
BJT



MOSFET

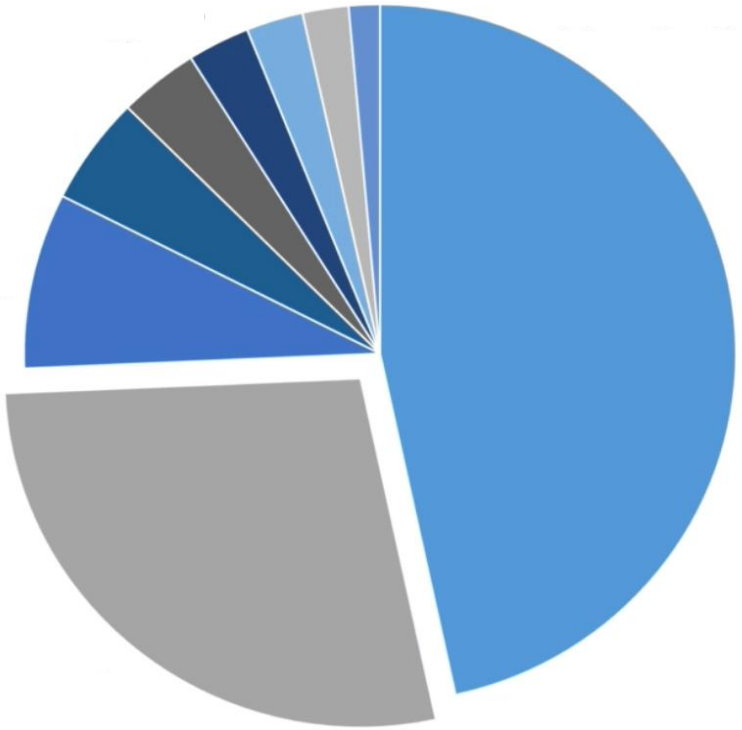


MOSFET (Complementary)



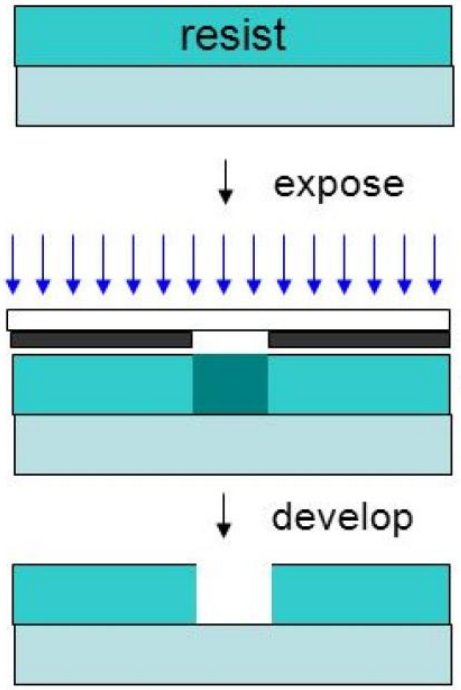
Wafer manufacturing

IN THE CRUST OF THE EARTH,



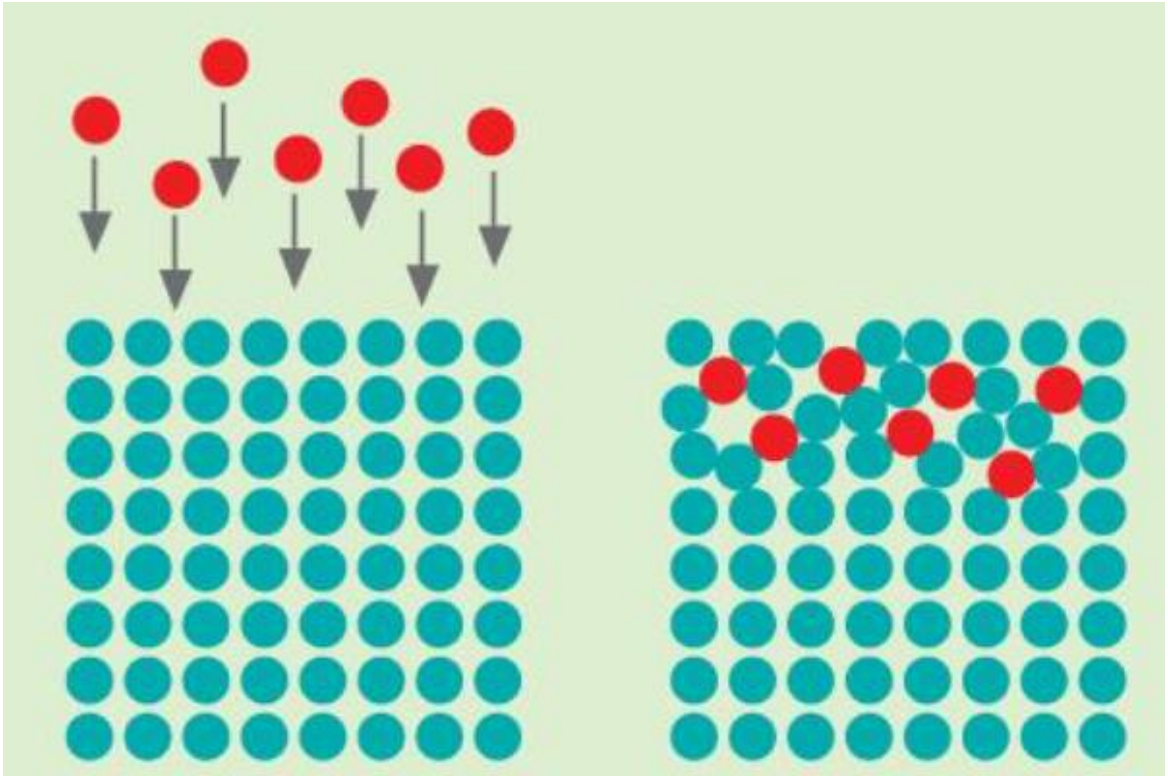
Photolithography

Photolithography uses optical radiation to image the mask on a silicon wafer using photoresist layers.



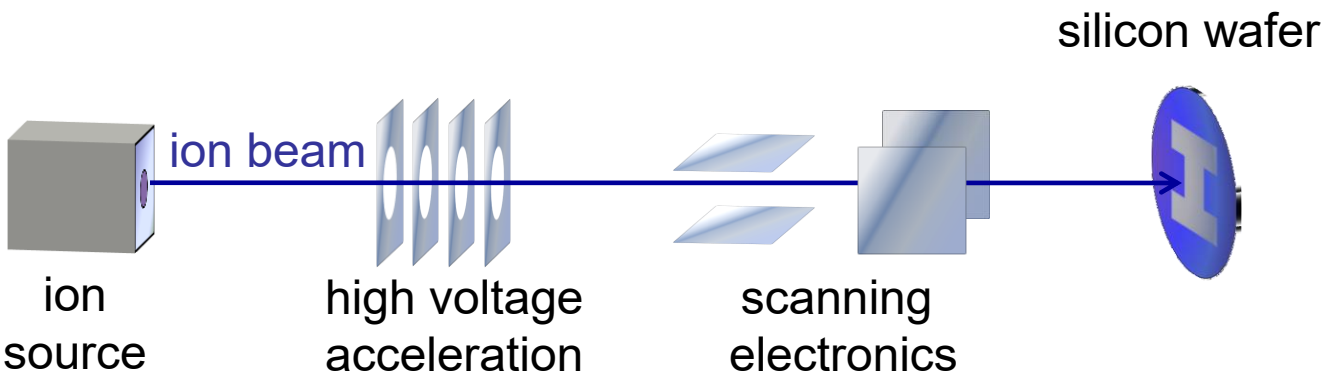
Doping methods: Diffusion & Implantation

Ion implantation



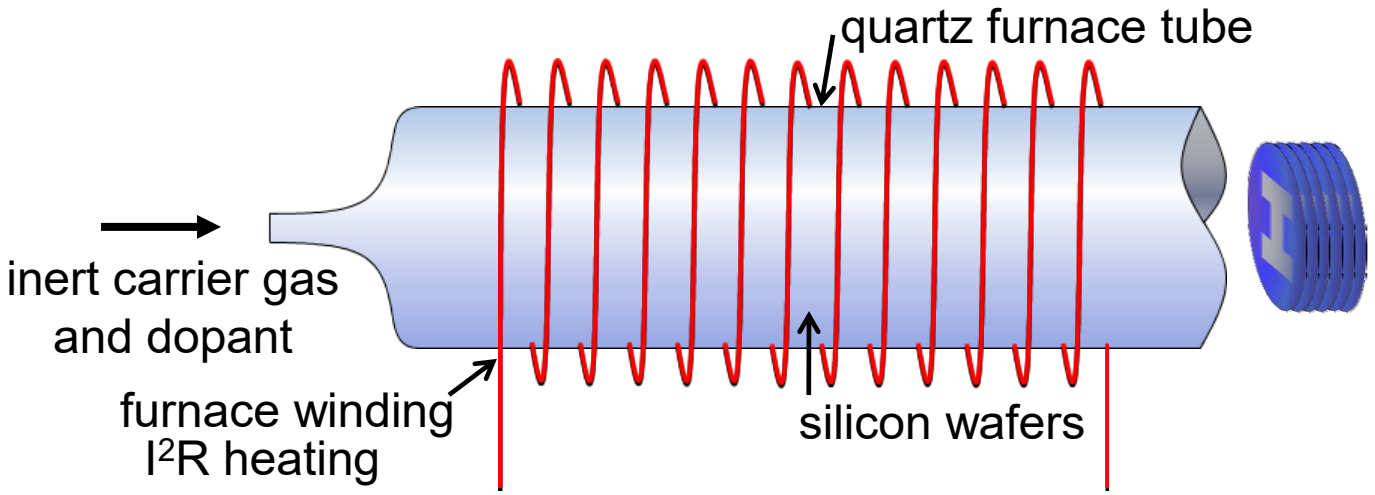
Doping methods: Diffusion & Implantation

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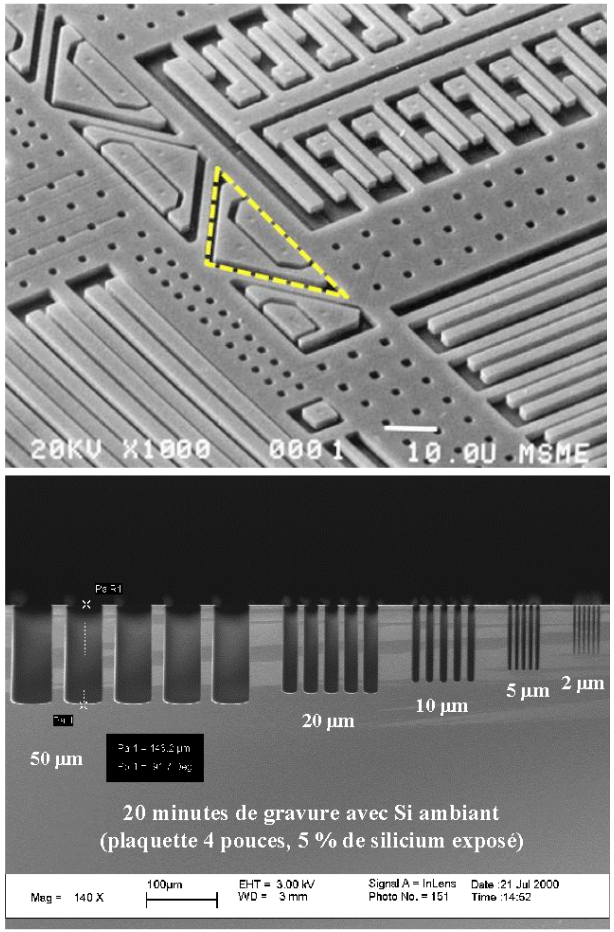
Doping methods: Diffusion & Implantation

Diffusion

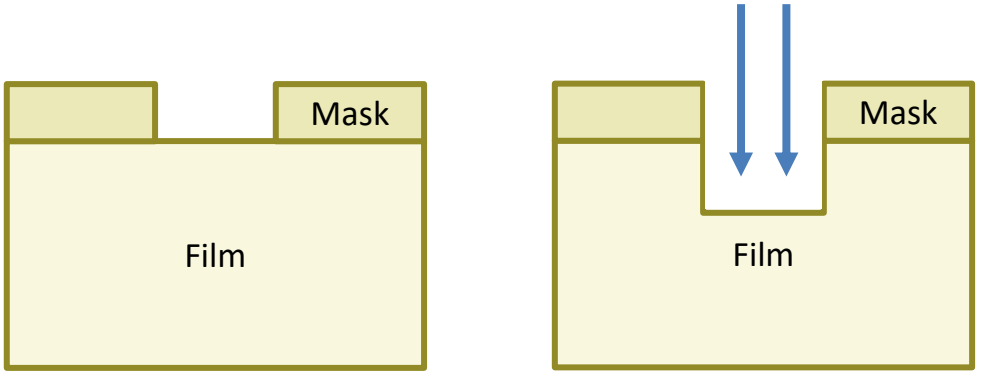


$$2B_2O_3 + 3Si \rightarrow 3SiO_2 + 4B$$
$$2P_2O_5 + 5Si \rightarrow 5SiO_2 + 4P$$

Etching: Removing materials



Selective removal using mask



Deposition (metal, dielectric)

